

Deep Reinforcement Learning

Offline Reinforcement Learning

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Content

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- Learning policies from datasets
- Offline reinforcement learning
 - Distributional shifts
 - Overestimation bias
 - Relation to imitation learning
 - Algorithmic evolution of methods
- Policy constraint methods
 - Forward and backward KL divergence
 - RL + BC (behavioral cloning)
 - advantage-weighted regression
 - Batch-constrained Q -learning

- Implicit Q -learning
- Conservative Q -learning
- Offline-to-online RL
- Model-based offline RL

Where are we?

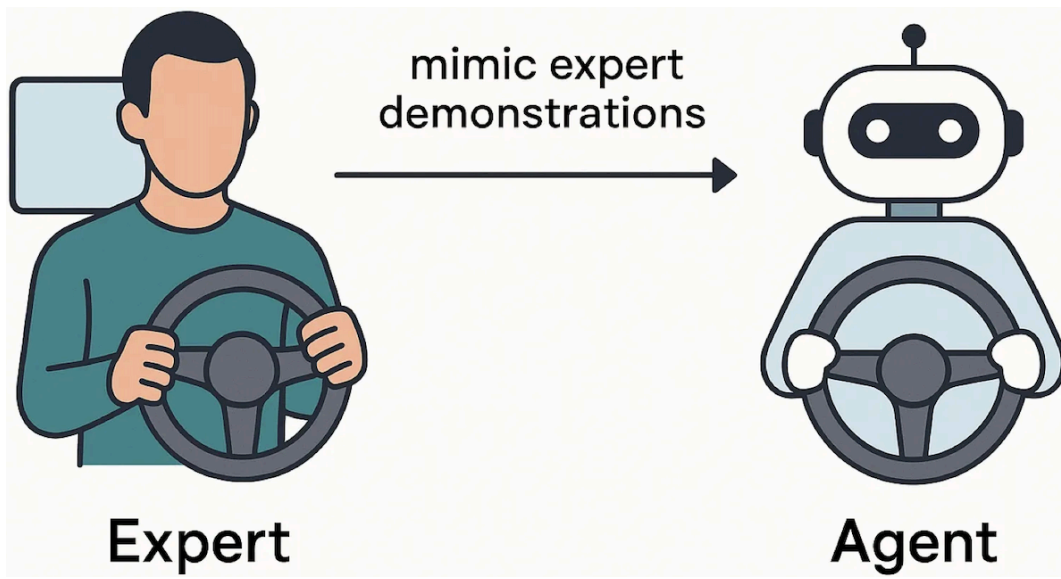
Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6-13	DQN, policy gradients, actor-critic, PPO/SAC, exploration	Deep RL from basics to modern algorithms
	Model-Based Control	
14-15	Optimal & feedback control, MPC, planning, model-based RL	Planning & control when we know the model
	Advanced Topics	
16	Offline reinforcement learning	RL with a static, given dataset
	Imitation learning	
	Transfer learning	

Lecture contents

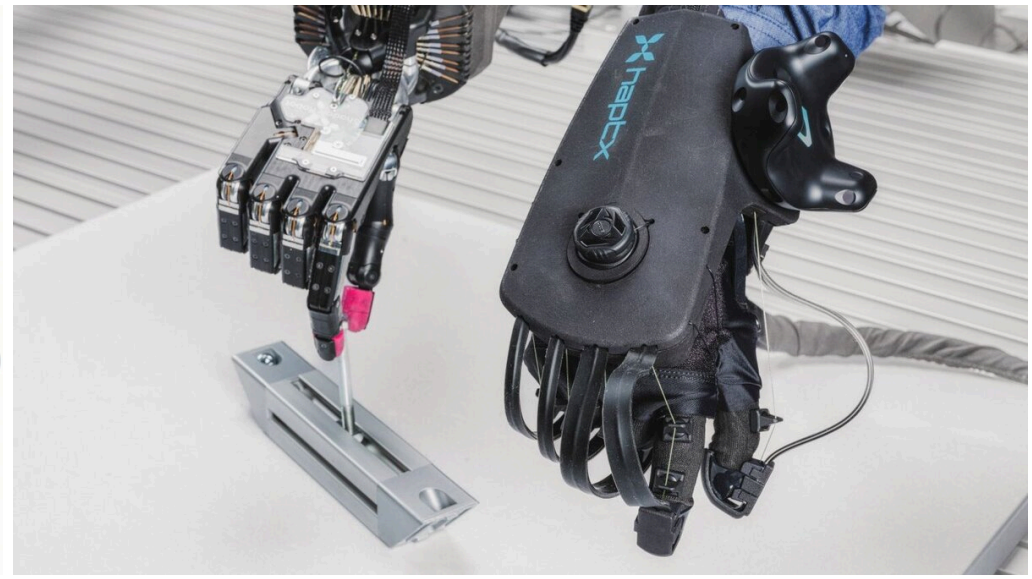
Learning policies from datasets

Imitation learning

- Two of the biggest challenges in RL:
 1. Long waiting times until we find a policy that does not fail completely.
 2. Designing good reward functions that actually make the agent do what we want it to do.
- What can we do to circumvent this?
⇒ “Let’s just show the agent how it’s done?”: learn from an expert!
- The concept behind this is **imitation learning**.
 - Given examples from a human expert, try to derive a policy that mimicks this behavior as closely as possible ...
 - ... while generalizing to previously unseen scenarios.



Source: [Medium.com](#).



Source: [Innovation campus future mobility](#).

Some developments in imitation learning (1)

Behavioral cloning (BC)

- Treats imitation as **supervised learning**. Inputs: expert states; Outputs/Labels: expert actions as labels
- Ignores the underlying MDP.

+ Simple to implement.

+ No environment interaction during training.

+ Very fast.

— Compounding errors / covariate shift: A small error at step one puts the agent in a state it has never seen before, causing increasingly poor decisions.

— Assumes i.i.d. data distribution, which is false in sequential environments.

Inverse reinforcement learning (IRL)

- Instead of copying the actions, IRL tries to **figure out the intent** by assuming that the expert is optimizing an unobserved reward function.
- Extracts hidden reward function from the demonstrations and then runs standard RL on top.

+ Robust to compounding errors.

+ Reward function can transfer well even if the environment changes slightly.

— Expensive inner-loop problem: RL training in each iteration.

— Mathematically underdetermined (many different reward functions can explain the same expert behavior).

Some developments in imitation learning (2)

Interactive / Human-in-the-Loop IL

- To fix BC's compounding errors without the cost of IRL, methods like *Dataset Aggregation* (DAgger, (Ross, Gordon, and Bagnell 2011)) were introduced.
 - Runs current policy in the environment.
 - When it makes a mistake and goes off-track, an expert intercepts and provides the correct action labels for those erroneous states.
 - This data is added back into the training set.

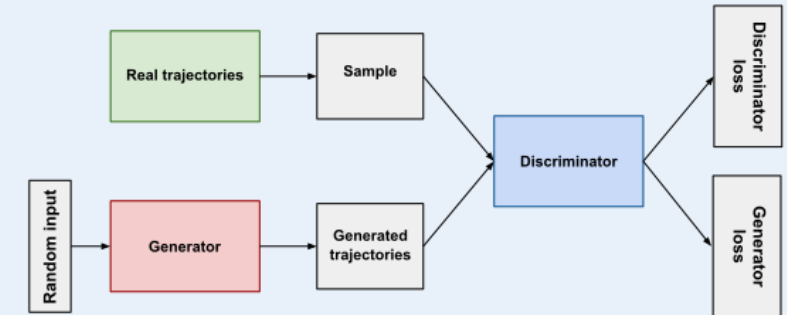
+ Directly fixes covariate shift.

+ forces the agent to learn how to recover from its own mistakes.

— Requires an expert to be constantly available and online during training to label data.

Generative adversarial imitation learning (GAIL) (Ho and Ermon 2016)

- Two-player game:
 - Generator: the agent trying to mimic the expert policy.
 - Discriminator: classifier trying to distinguish between agent and expert trajectories.



Inspired by Generative Adversarial Networks (GANs).

- The discriminator penalizes the agent when it does not create expert-like trajectories.

+ Mimicks entire trajectories, not just single transitions as BC.

+ No expensive inner loop as in IRL.

+ Does not require an interactive expert like DAgger.

— Inherits GAN instabilities (mode collapse, highly sensitive hyperparameters).

- Exhausting for humans and frequently impossible for complex setups.

- Very data hungry.

What if we want to do better than the data?

Algorithm (class)	Key concept
Behavioral cloning (BC)	Copy data (i.e., “expert” as closely as possible).
Inverse RL (IRL)	Identify reward from data $\Rightarrow$ then standard RL.
Dataset aggregation (DAgger)	Expert corrects faulty behavior (meaning there is an online component).
Generative adversarial imitation learning (GAIL)	The generative version of BC mimicking trajectories.

The key ideas in imitation learning

BUT: What if we cannot get new data, but we still want to be better than the performance of the dataset?

Maybe the data came from

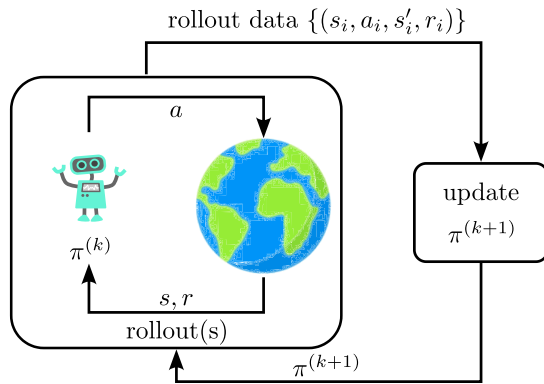
- Experts,
- semi-experts,
- one or more past RL policies,
- a mixture of the above?

$\Rightarrow$ **Offline reinforcement learning!**

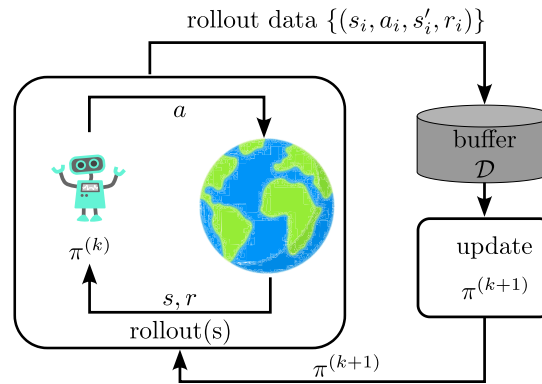
Offline Reinforcement Learning

What is offline RL?

On-policy RL

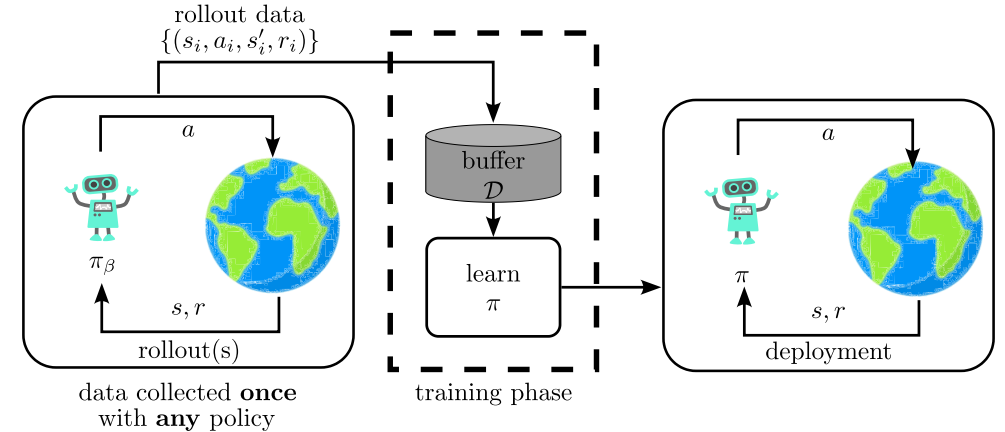


Off-policy RL



Inspired by (Levine et al. 2020).

Offline reinforcement learning



Offline RL is closely related to supervised learning / data-driven AI!

- Data-driven AI is about learning from large datasets.
 - + Learns about the real world from data.
 - Doesn't try to do better than the data.
- Reinforcement learning is about optimization.
 - + Optimizes a goal with emergent behavior.
 - Doesn't make use of real-world data.

⇒ Data without optimization doesn't allow us to solve new problems in new ways.

⇒ Optimization without data is hard to apply to the real world outside of simulators.

Offline reinforcement learning: Use *prior data* for RL training, *without any interactions* with the environment!

Goal: Given a dataset $\mathcal{D}$, learn the best possible policy π_ϕ that is *supported by the dataset*.

Can offline RL work?

1. Find the “good stuff” in a dataset full of good and bad behaviors.

Example, consider a dataset of various drivers:

- with imitation learning, we would find the average performance 🙅
- with offline RL, we hope to extract a policy mimicking the best driver’s performance 👍

2. **Generalization**: good behavior in one place may suggest good behavior in another place.

Driving example: we improve upon the best driver by generalizing to unseen situations.

3. **“Stitching”**: parts of good behaviors can be recombined.

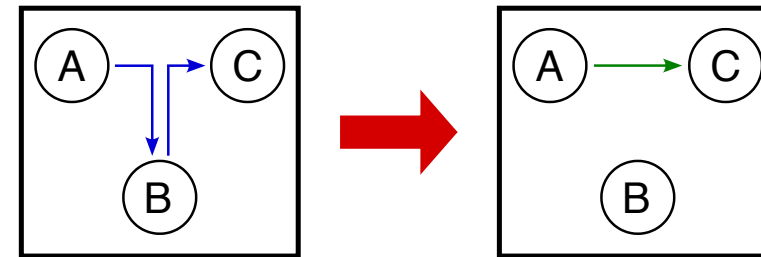
Takeaway: It’s not just imitation learning!

Intuition: Get order from chaos.

- On micro scales (e.g., a “straight path” from a bunch of wiggly trajectories).
- On macro scales (e.g., $A \rightarrow B \rightarrow C \Rightarrow A \rightarrow C$).



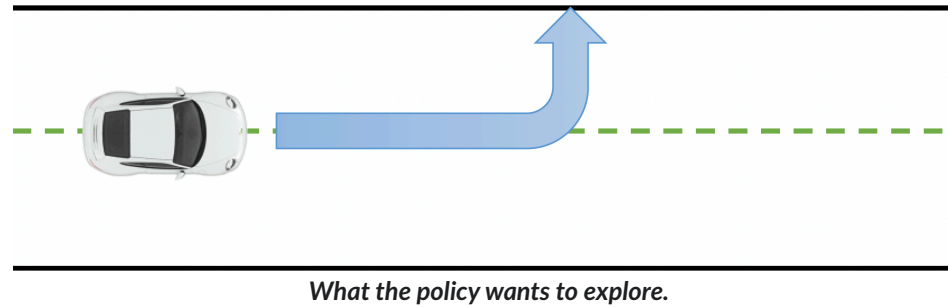
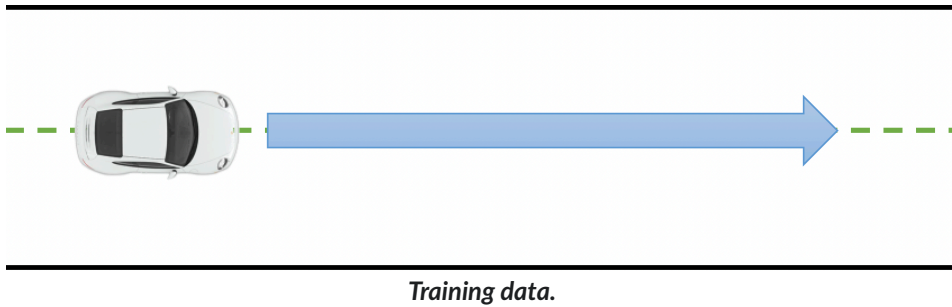
Generalization (Source: [D4RL](#)).



Stitching (Source: Sergey Levine’s [CS285 lecture](#)).

Why is offline RL so hard?

Fundamental problem: counterfactual queries



- Is this good or bad?
- How can we know we have never seen it?

- **Online RL** algorithms don't have to handle this.
 - They can simply try this action and see what happens.
- **Offline RL** methods have to account for these unseen (OOD: "out of distribution") actions,
 - ideally in a safe way ...
 - while still making use of generalization to come up with behaviors that are better than the best thing seen in the data!

Example: Q-learning / Bellman update

$$Q_{\theta} \leftarrow r + \gamma \mathbb{E}_{s' \sim p(s'|s,a)} \left[\underbrace{\max_{a' \in \mathcal{A}} Q_{\bar{\theta}}(s', a')}_{\text{can go OOD}} \right]$$

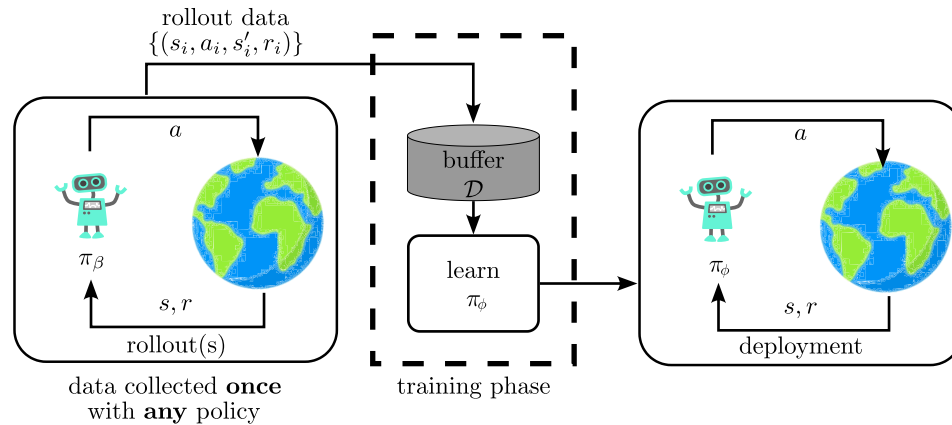
💡 *out of distribution vs. out of sample.*

- Out of sample $\Rightarrow$ generalization: Evaluation outside the training dataset, but on data **from the same distribution**.

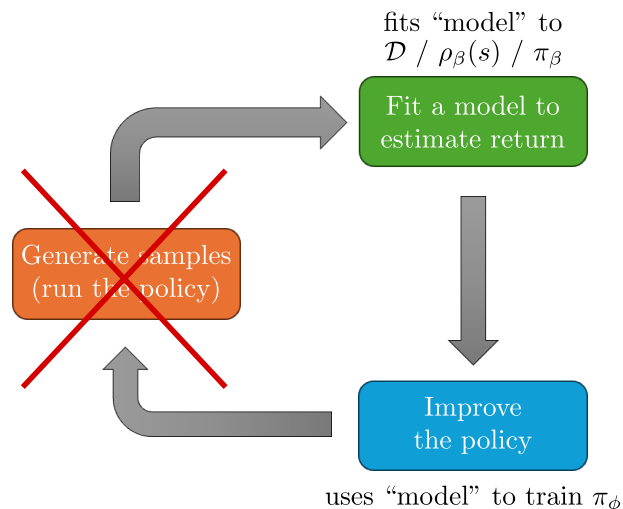
- Out of distribution $\Rightarrow$ Evaluation on unseen **data from a different distribution**. This is much harder!

Distributional shifts (1)

Offline reinforcement learning



Inspired by (Levine et al. 2020).



Inspired by Sergey Levine's CS285 lecture.

Formally:

$$\begin{aligned} \mathcal{D} &= \{(s_i, a_i, s'_i, r_i)\}_{i=1}^N \\ s &\sim \rho_\beta(s) \\ a &\sim \pi_\beta(a|s) \quad (\text{generally unknown}) \\ s' &\sim p(s'|s, a) \end{aligned}$$

RL objective: $\max_{\phi} \sum_{t=0}^T \mathbb{E}_{s \sim \rho_\phi, a \sim \pi_\phi(\cdot|s)} [\gamma^t r_t]$

- "Model" is valid under π_β ...
- **not** under π_ϕ !
 - "Model" could be the Q -function Q_{π_ϕ} ,
 - or the dynamics $p_\phi(s'|s, a)$.

Distributional shifts (2)

Example: Empirical risk minimization (maximum likelihood)

$$\theta = \arg \min_{\hat{\theta}} \mathbb{E}_{x \sim p(x), y \sim p(y|x)} \left[\left(f_{\hat{\theta}}(x) - y \right)^2 \right]$$

Question: Given some x^* , is $f(x^*)$ correct?

- $\mathbb{E}_{x \sim p(x), y \sim p(y|x)} \left[\left(f_{\hat{\theta}}(x) - y \right)^2 \right]$ is low

$\Rightarrow$ probably yes.

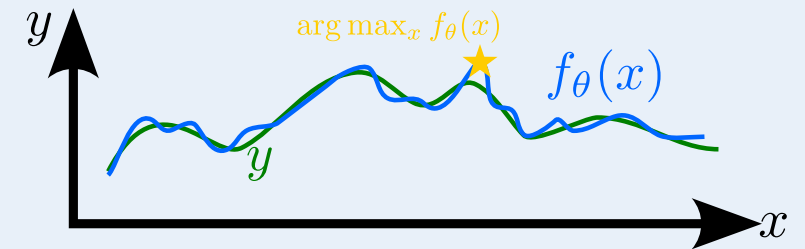
- $\mathbb{E}_{x \sim \bar{p}(x), y \sim p(y|x)} \left[\left(f_{\hat{\theta}}(x) - y \right)^2 \right]$ is generally not, if $\bar{p}(x) \neq p(x)$

$\Rightarrow$ probably no.

- What if $x^* \sim p(x)$?
 - Not necessarily...
 - but neural networks tend to generalize well!

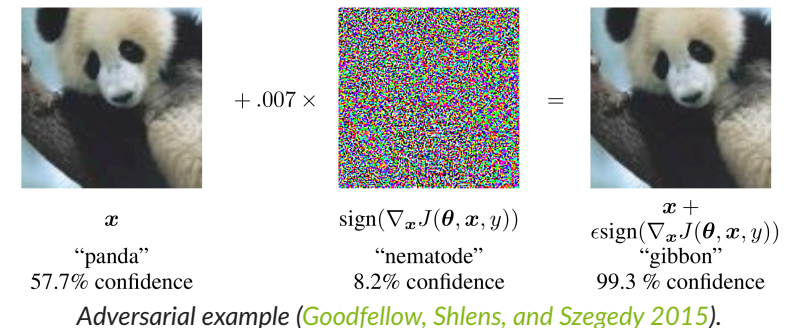
What if we pick the maximizing x^* ?

$$x^* = \arg \max_x f_{\theta}(x)$$



Inspired by (Levine et al. 2020).

💡 This is exactly what we're exploiting in adversarial learning!



Overestimation bias in offline RL

Q-learning and Q-function actor-critic

$$\phi = \arg \min_{\hat{\theta}} \mathbb{E}_{s \sim \rho_{\beta}, a \sim \pi_{\beta}(\cdot|s)} [\|Q_{\theta}(s, a) - y\|^2],$$

$$y_i = r_i + \gamma \mathbb{E}_{s'_i \sim \rho_{\beta}, a' \sim \pi_{\phi}(\cdot|s'_i)} [Q_{\theta}(s'_i, a')]$$

- Trying to find the best policy: we want $\pi_{\phi}(a|s) \neq \pi_{\beta}(a|s)$!
- But we only have s'_i for s_i, a_i from our offline dataset.
- **Even worse:** We're actually picking an *adversarial example*,

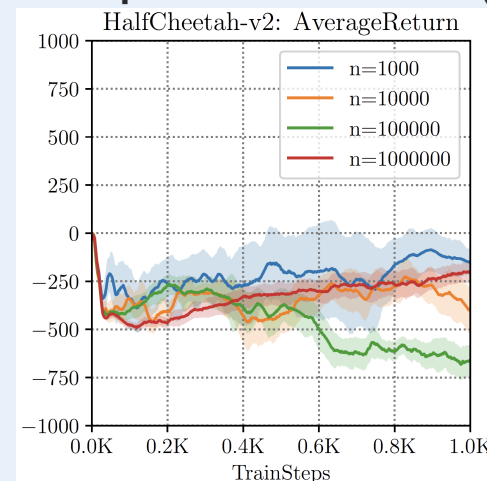
$$\phi = \arg \max_{\hat{\phi}} \mathbb{E}_{a \sim \pi_{\hat{\phi}}(\cdot|s)} [Q_{\theta}(s, a)].$$

Model-based RL

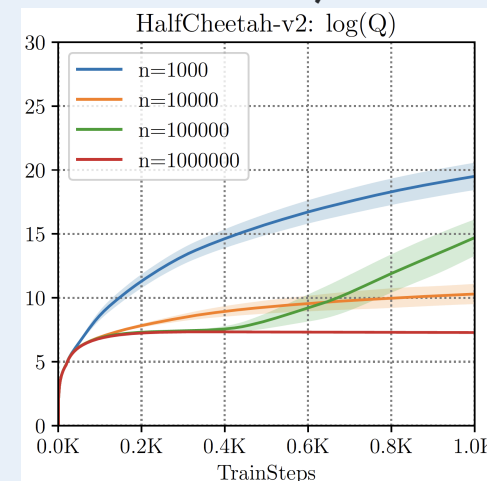
$$f = \arg \min_{\hat{f}} \mathbb{E}_{s \sim \rho_{\beta}, a \sim \pi_{\beta}(\cdot|s), s' \sim p(s'|s, a)} [\|f(s, a) - s'\|_2^2]$$

- Probably large: $\mathbb{E}_{s \sim \rho_{\beta}, a \sim \pi_{\phi}(\cdot|s), s' \sim p(s'|s, a)} [\|f(s, a) - s'\|_2^2]$.
- **Even worse:** Pick π_{ϕ} to maximize the reward under f !

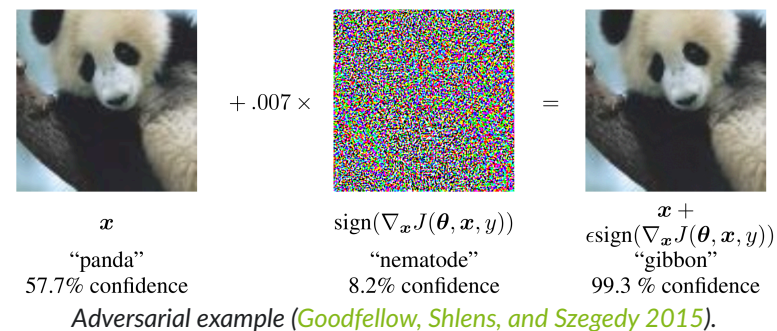
Example: SAC offline RL (Kumar et al. 2019)



How well the agent performs.



How well it thinks it does.



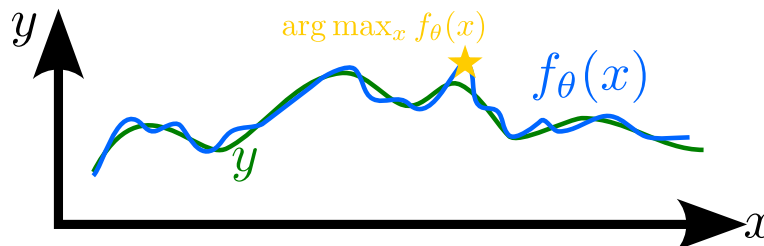
Algorithmic evolution of methods

The following strategies (or classes of approaches) are the most common ones today:

1. **Policy constraint methods:** Stay close to the data, e.g.,

$$D_{\text{KL}}(\pi_{\phi}(\cdot|s) \parallel \pi_{\beta}(\cdot|s)) \leq \epsilon.$$

2. **Value-regularization & pessimism:** Go anywhere, but assume what you haven't seen is dangerous $\Rightarrow$ avoid overestimation.



Inspired by (Levine et al. 2020).

3. **Implicit Q-learning:** Avoid out-of-distribution actions in updates.

Common theme: “Maximize return while staying close to the dataset.”

Policy constraint methods

Adding constraints

- Simplest approach to avoid overestimation: constrain the distance between behavior policy π_β and the learned policy π_ϕ .
- Remember the KL divergence?

$$D_{\text{KL}}(\pi_\beta(\cdot|s) \parallel \pi_\phi(\cdot|s)) = \mathbb{E}_{a \sim \pi_\beta(\cdot|s)} \left[\log \frac{\pi_\beta(a|s)}{\pi_\phi(a|s)} \right] = \mathbb{E}_{a \sim \pi_\beta(\cdot|s)} [\log \pi_\beta(a|s) - \log \pi_\phi(a|s)].$$

- Constraining the distance yields a constrained optimization problem:

$$\phi \leftarrow \arg \max_{\phi} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi(\cdot|s)} [Q(s, a)] \quad \text{s.t.} \quad D_{\text{KL}}(\pi_\beta(\cdot|s) \parallel \pi_\phi(\cdot|s)) \leq \epsilon. \quad (1)$$

- Where have we seen this before? $\Rightarrow$ Natural policy gradient!
- Via **Lagrange multipliers**: additional term in the actor loss function:

$$\begin{aligned} \phi &\leftarrow \arg \max_{\phi} \mathbb{E}_{s \sim \mathcal{D}} [\mathbb{E}_{a \sim \pi_\beta(\cdot|s)} [Q(s, a)] - \lambda D_{\text{KL}}(\pi_\beta(\cdot|s) \parallel \pi_\phi(\cdot|s))] \\ \Rightarrow \quad \phi &\leftarrow \arg \max_{\phi} \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\beta(\cdot|s)} [Q(s, a) + \lambda \log \pi_\phi(a|s) + \text{const}]. \quad (2) \end{aligned}$$

- To solve (1), we need to solve for both ϕ and the Lagrange multiplier λ .

AC + BC methods

Methods of the type (2) are often termed “AC+BC”:
actor-critic plus behavior cloning,
as the term $\log \pi_\phi(a|s)$ is typical in BC methods. Popular examples:

- Alternative: treat λ as a hyperparameter.

- TD3 + BC,
- SAC + BC.
- DDPG + BC,

Forward KL versus reverse KL

- Remember: the KL divergence is not a real **metric**. In particular, it violates the symmetry condition, i.e.,

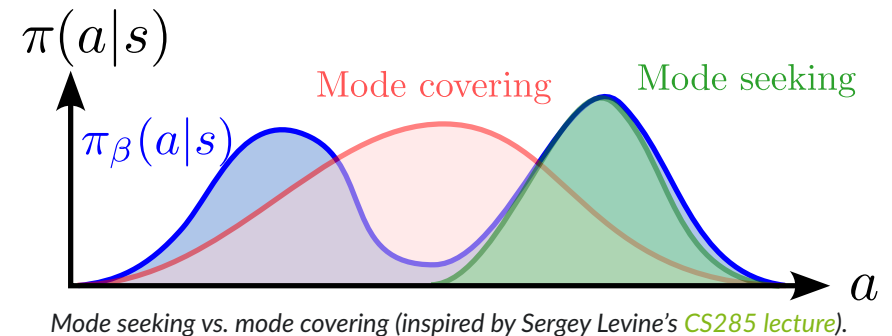
$$D_{\text{KL}}(\pi_{\phi}(\cdot|s) \parallel \pi_{\beta}(\cdot|s)) = \mathbb{E}_{a \sim \pi_{\phi}(\cdot|s)} \left[\log \frac{\pi_{\phi}(a|s)}{\pi_{\beta}(a|s)} \right] \neq \mathbb{E}_{a \sim \pi_{\beta}(\cdot|s)} \left[\log \frac{\pi_{\beta}(a|s)}{\pi_{\phi}(a|s)} \right] = D_{\text{KL}}(\pi_{\beta}(\cdot|s) \parallel \pi_{\phi}(\cdot|s)).$$

- Forward KL** a.k.a. “mode covering”:

$$D_{\text{KL}}(\pi_{\beta}(\cdot|s) \parallel \pi_{\phi}(\cdot|s)) = \mathbb{E}_{a \sim \pi_{\beta}(\cdot|s)} [\log \pi_{\beta}(a|s) - \log \pi_{\phi}(a|s)].$$

- Reverse KL** a.k.a. “mode seeking”:

$$\begin{aligned} D_{\text{KL}}(\pi_{\phi}(\cdot|s) \parallel \pi_{\beta}(\cdot|s)) &= \mathbb{E}_{a \sim \pi_{\phi}(\cdot|s)} [\log \pi_{\phi}(a|s) - \log \pi_{\beta}(a|s)] \\ &= -\mathbb{E}_{a \sim \pi_{\phi}(\cdot|s)} [\log \pi_{\beta}(a|s)] - \mathcal{H}(\pi_{\phi}(\cdot|s)). \end{aligned}$$



- Mode covering: What happens if for some (s, a) ,
 $\pi_{\phi}(a|s) = 0$ while $\pi_{\beta}(a|s) > 0$?
 - KL divergence 💥 $\Rightarrow$ Need to cover everything from the dataset!
- Mode covering: What happens if for some (s, a) ,
 $\pi_{\beta}(a|s) = 0$ while $\pi_{\phi}(a|s) > 0$?

Which one is desirable?

- Mode seeking is what we actually want.
- Extract the best behavior from a dataset.

◦ KL divergence 🌟 $\Rightarrow$ Need to ensure π_ϕ stays within a subset of π_β .

- However, it is harder to realize, which is why we often use mode covering in practice, see $AC + BC$.

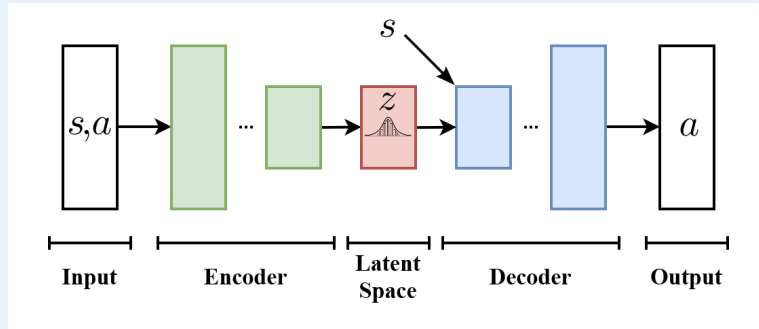
How to learn π_β if we don't know it

- Constraining the KL divergence requires knowledge of the behavior policy π_β .
- What can we do if we don't know it? $\Rightarrow$ extract it from the dataset!

Behavior cloning.

- Given a dataset $\mathcal{D} = \{(s_i, a_i)\}_{i=1}^N$, we have a probabilistic supervised learning problem.
- Find the probability distribution $\pi_\psi(a|s)$ giving rise to $\mathcal{D}$.
- In other words maximize the likelihood of a_i given s_i over $\mathcal{D}$.
- Approaches:

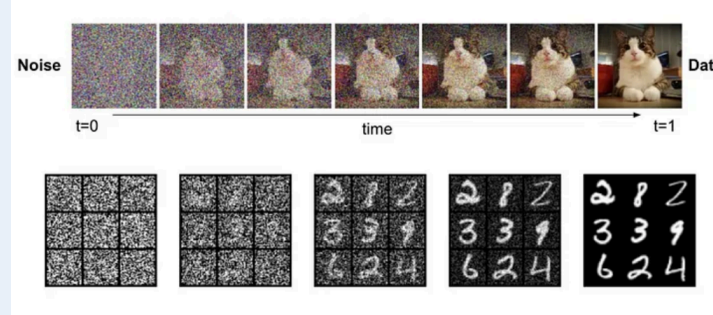
- Variational autoencoder (VAE) (Kingma and Welling 2022):



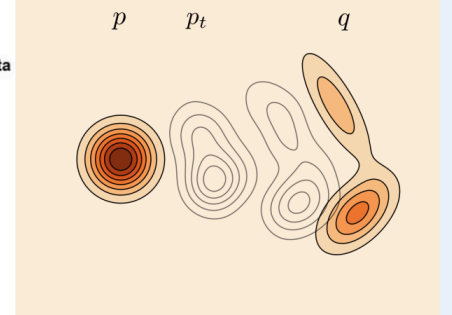
Adapted from Wikipedia.

- Gaussian policy: $\pi_{\psi}(a|s) = \mathcal{N}(\mu_{\psi}(s), \sigma_{\psi}^2(s))$.

- Generative modeling



Source: Medium.com.



Alternative to KL divergence

- The KL divergence is a natural choice for the policy constraint, but not necessarily the best one:
 - + Mathematically tractable / closed-form solutions.
 - + Direct penalization of OOD actions.
 - + Smooth/soft regularization with tunable hyperparameter λ .
 - Asymmetry (forward KL vs. reverse KL).
 - Overly conservative in unseen regions.
 - Sensitive to behavior policy estimation.
- More generally, **we do not even want the two distributions to match!**
- We just want to find the best behavior supported by the dataset.
- Alternative: Minimizing **Maximum Mean Discrepancy (MMD)**, see (Kumar et al. 2019):

$$MMD^2(\pi, \beta) = \mathbb{E} [k(a_\pi, a'_\pi)] - 2\mathbb{E} [k(a_\pi, a_\beta)] + \mathbb{E} [k(a_\beta, a'_\beta)].$$

- k is a kernel function for implicitly evaluating inner products between feature vectors: $k(a_\pi, a_\beta) = \psi(a_\pi)^\top \psi(a_\beta)$.
- $\mathbb{E}[k(a_\pi, a'_\pi)]$: How close to each other are the actions generated by our policy?
- $-2\mathbb{E}[k(a_\pi, a_\beta)]$: How close are our actions to the dataset actions?
- $\mathbb{E}[k(a_\beta, a'_\beta)]$: The internal variance of the dataset actions (a constant baseline).

- + Bypasses density estimation / works on samples .
- + No infinite penalty on unsupported actions.
- + Allows for multimodel generalization.
- Computationally expensive.
- High sample variance / hard to tune the kernel bandwidth.
- Curse of dimensionality.

The BRAC framework

Unification of various approaches in the BRAC framework

In (Wu, Tucker, and Nachum 2019), the authors present the *Behavior Regularized Actor-Critic (BRAC)*.

- This can be seen as a unification of many methods.
- The constraint can be enforced in two different ways
 - In the policy approximation (*Policy Regularization*)
 - In the value or Q function approximation (*Value Regularization*)
- The constraint is defined as a general distance metric.
 - The KL divergence and MMD are then special cases.
- Additional design choice: How to approximate the expectation.

Implicit Q -learning

Advantage-weighted regression

(Peng et al. 2019)

Implicit Q -learning (IQL)

Value-regularization & pessimism

Conservative Q -learning

Model-based offline RL

Learning a model, then RL

Offline-to-online RL

A straightforward engineering approach

(Ball et al. 2023)

Benchmarking

The D4RL benchmark

<https://sites.google.com/view/d4rl-anonymous/>.

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